# Abstracts Lie?

*Measuring Thematic Alignment Between Journal Scope and Article Abstracts*

*Case study: Journal of Machine Learning Research (JMLR)*

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# Research Question

*How reliably do article abstracts represent the thematic focus declared in a journal's Aims & Scope?*

## AIMS & SCOPE
Journal-level thematic description

## ABSTRACTS
Paper-level summaries

## GOAL
Measure alignment between them

# *DATASET

JMLR

"The Journal of Machine Learning Research (JMLR), established in 2000, provides an international forum for the electronic and paper publication of high-quality scholarly articles in all areas of machine learning."

## 80 PAPER ABSTRACTS

TITLE | ABSTRACT | YEAR | URL

# PROJECT PIPELINE

```
src
  data_collection
    collect_papers.py
  evaluation
    compare.py
    outliers.py
  preprocessing
    clean_text.py
  representation
    embedding.py
    tfidf.py
  visualization
    plots.py
```

JMLR website

Collect abstracts

*collect_papers.py*

Clean Text

*clean_text.py*

TF-IDF vectors

*tfidf.py*

Embedding vectors

*embedding.py*

Cosine Similarity

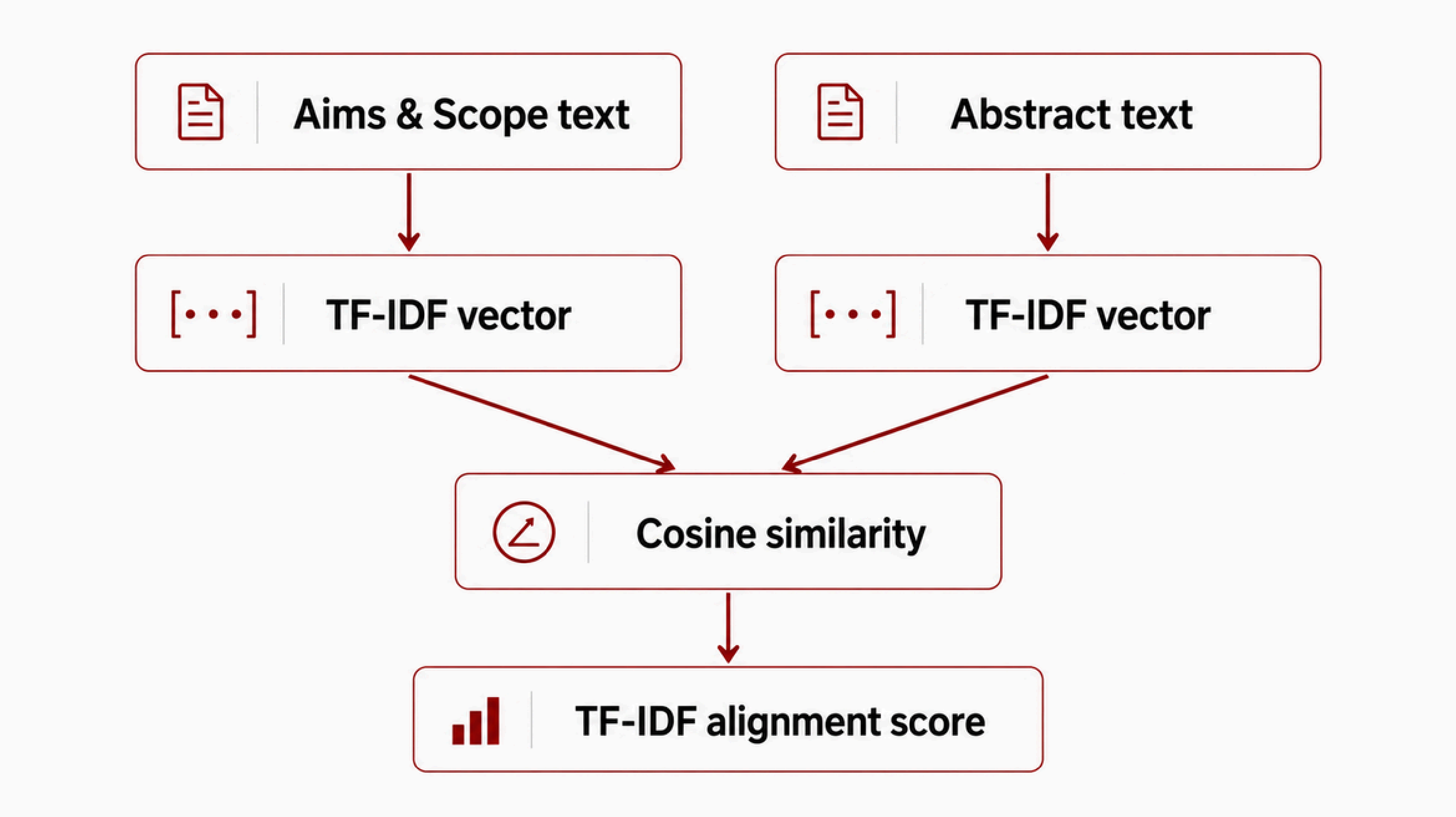
Comparison. Outliers. Plots.

*compare.py | outliers.py | plots.py*

# TF-IDF Method

## Lexical baseline

- Converts text into sparse word-based vectors
- Measures vocabulary overlap
- Uses cosine similarity




```
vectorizer = TfidfVectorizer(
    lowercase=True,
    stop_words="english",
    max_features=5000
)
tfidf_matrix = vectorizer.fit_transform(texts)
aims_vector = tfidf_matrix[0]
abstracts_vectors = tfidf_matrix[1:]
scores = cosine_similarity(abstracts_vectors, aims_vector).flatten()
```

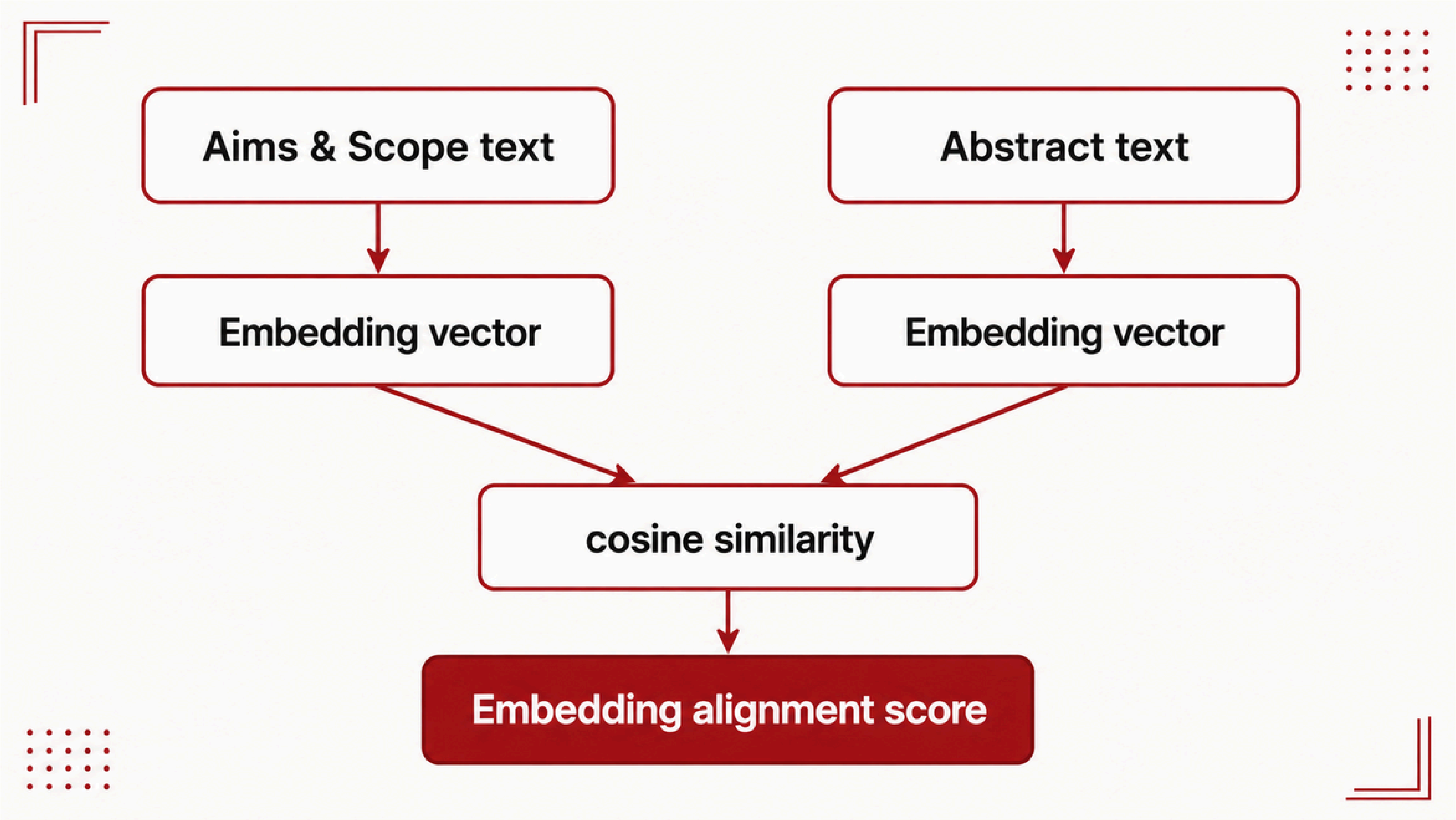
# Embedding Method

## Semantic comparison

***Embeddings represent text as dense numerical vectors that capture semantic meaning.***

## *Why used?*

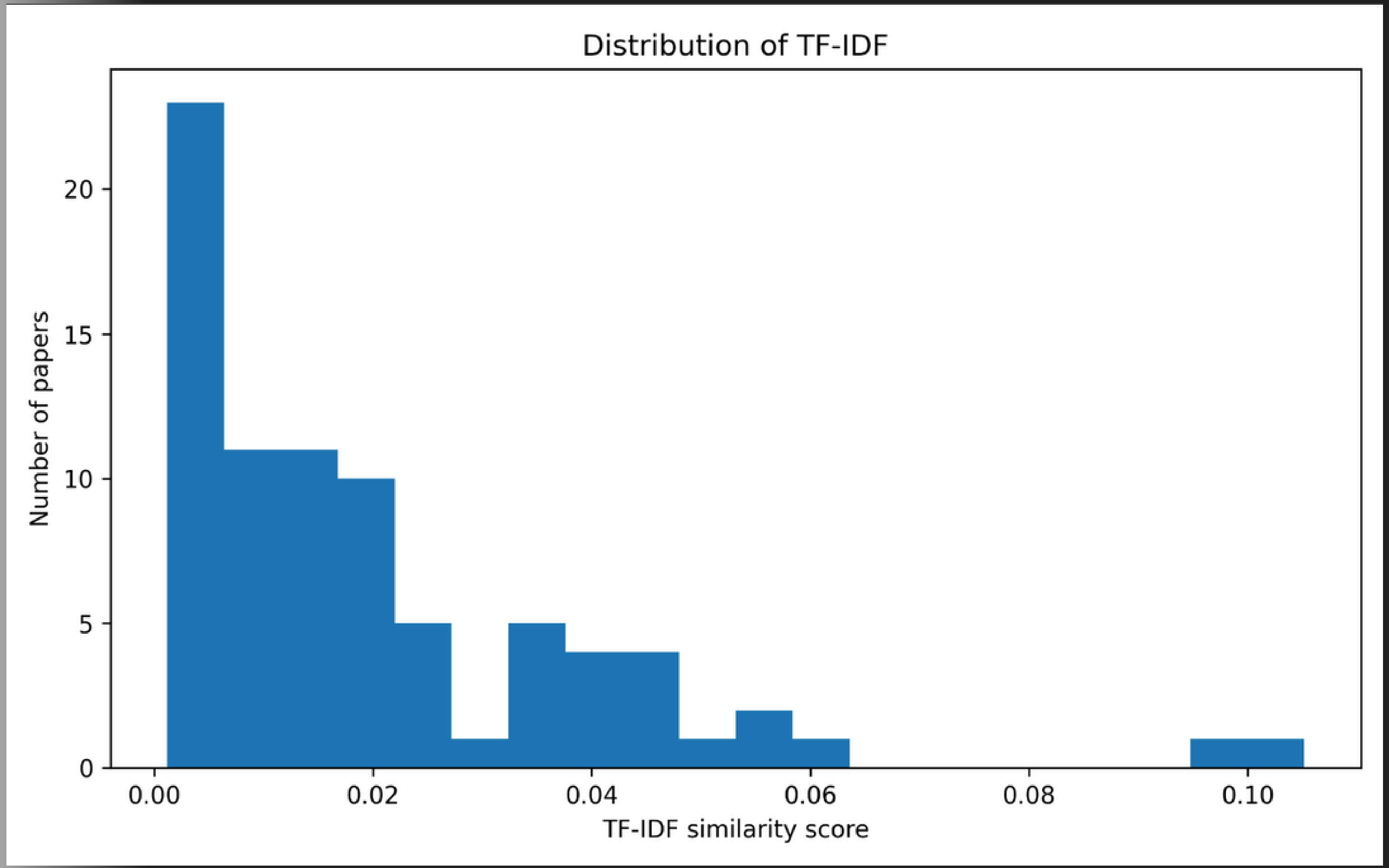
They can detect similarity even when two texts use different words



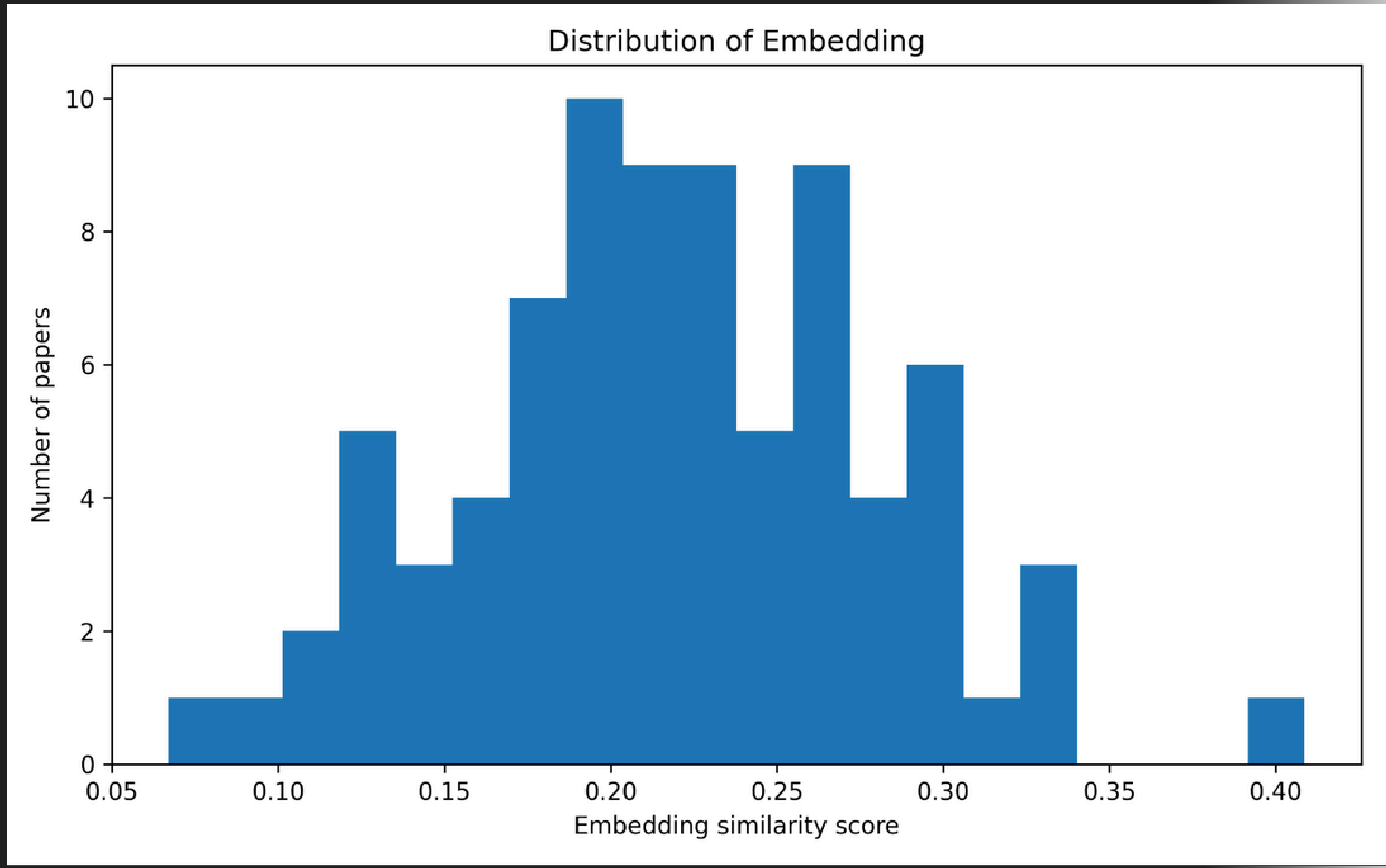
# Main Results

| Method | Mean | Min | Max |
|---|---|---|---|
| TF-IDF | 0.020 | 0.001 | 0.105 |
| Embeddings | 0.219 | 0.067 | 0.409 |

*"Embedding similarity is much higher than TF-IDF similarity"*




*TF-IDF scores are low because abstracts use technical vocabulary.*



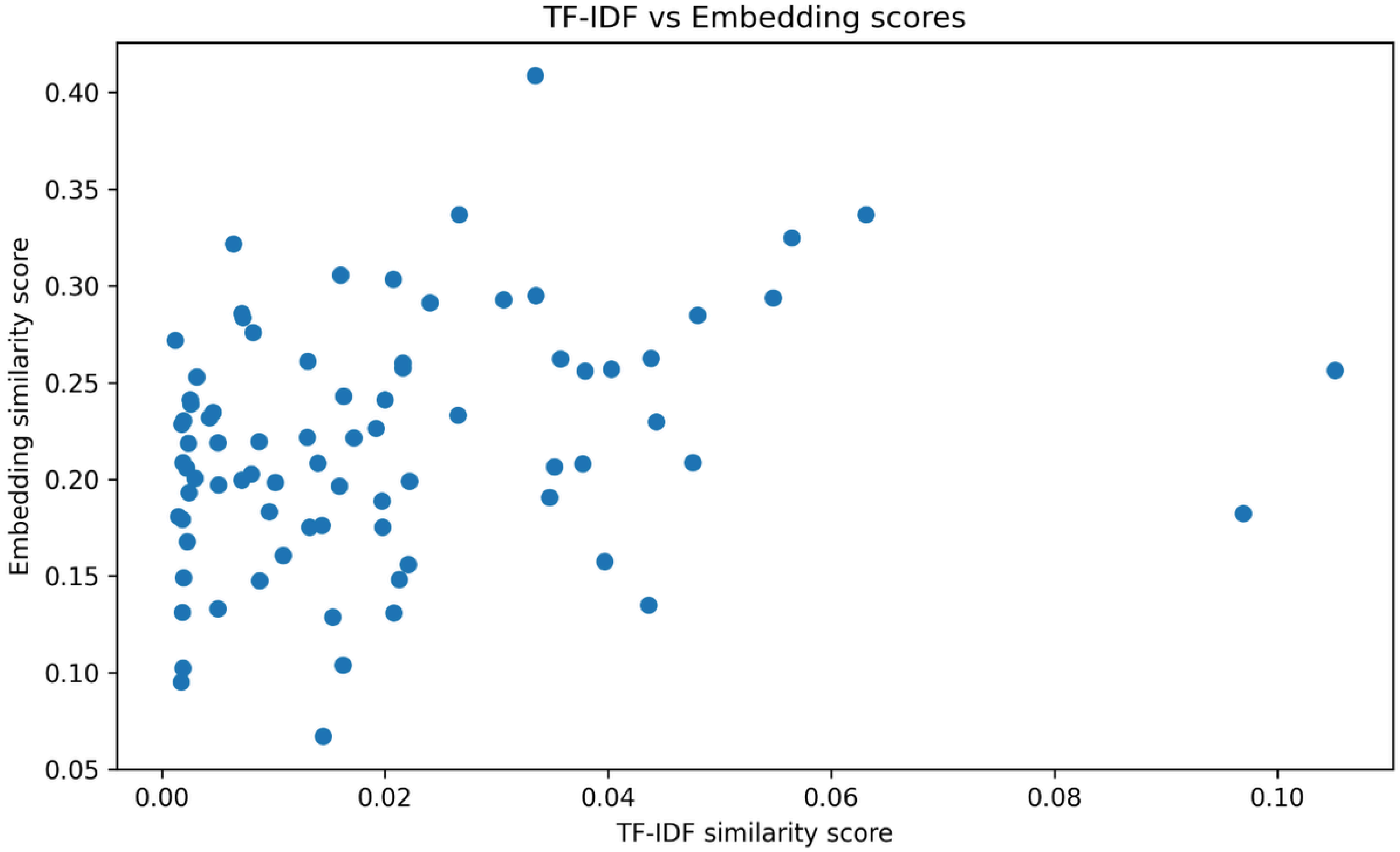

*Embedding scores are higher because embeddings capture semantic similarity.*

# Method Comparison

*TF-IDF and embeddings measure different types of alignment.*

***Correlation = 0.279***

Weak positive relationship




***TF - IDF***

Lexical overlap

Embeddings

Semantic similarity

*The methods partly agree, but they do not rank abstracts in the same way.*

# Outlier Analysis

*Low allignment does not automatically mean irrelevant*

## ✳ LOW SCORE ≠ IRRELEVANT

***Lowest embedding alignment papers were mostly about:***

- Semismooth Newton methods
- Unlabeled PCA and matrix methods
- Sketching operators
- Linear dynamical systems
- Nonconvex ADMM

| Paper type | Why low alignment? |
|---|---|
| Optimization | technical math vocabulary |
| Matrix methods | specialized terminology |
| Dynamical systems | not explicitly “machine learning” |

# Limitations

- Aims & Scope is short and general
- Dataset contains 80 abstracts
- Abstracts are not full papers
- Embedding model is pretrained, not domain-specific

# Conclusion

- Abstracts are useful, but imperfect thematic proxies
- TF-IDF underestimates alignment
- Embeddings capture broader semantic similarity
- Results are exploratory